

# Complete NLP Roadmap

## Complete NLP Roadmap (Beginner → Advanced → Research/Production)

Natural Language Processing (NLP) = teaching machines to understand, process, generate, and reason with human language.

### PHASE 0 — Prerequisites

- Python for NLP
- NumPy, Pandas, Matplotlib
- Linear Algebra
- Probability & Statistics
- Calculus
- Information Theory

### PHASE 1 — Foundations of NLP

- NLP Overview
- Text Cleaning
- Tokenization
- Stemming
- Lemmatization
- Stopwords
- Regular Expressions
- Linguistics for NLP

### PHASE 2 — Traditional NLP

- Bag of Words
- N-grams
- TF-IDF
- Traditional ML Algorithms

### PHASE 3 — Word Embeddings

- Word2Vec
- GloVe
- FastText
- Cosine Similarity

## **PHASE 4 — Deep Learning for NLP**

- Neural Networks
- RNN
- LSTM
- GRU
- Seq2Seq Models
- Attention Mechanism

## **PHASE 5 — Transformers**

- Self Attention
- Multi-head Attention
- Positional Encoding
- Encoder vs Decoder
- BERT
- GPT
- T5

## **PHASE 6 — Large Language Models (LLMs)**

- GPT Architecture
- Tokenization
- Pretraining
- Fine-tuning
- LoRA / QLoRA
- RLHF
- Prompt Engineering
- Hallucination Handling

## **PHASE 7 — NLP Tasks**

- Text Classification
- Sentiment Analysis
- Named Entity Recognition
- POS Tagging
- Machine Translation
- Summarization
- Question Answering
- Chatbots
- Topic Modeling

## **PHASE 8 — Advanced NLP**

- RAG
- Vector Databases
- Multimodal NLP
- AI Agents
- Knowledge Graphs
- Speech NLP

## **PHASE 9 — NLP Evaluation**

- Accuracy
- Precision
- Recall
- F1 Score
- BLEU
- ROUGE
- LLM Evaluation

## **PHASE 10 — NLP Deployment**

- FastAPI
- Flask
- ONNX
- TensorRT
- Cloud Deployment
- MLOps

## **PHASE 11 — Research Topics**

- Diffusion Language Models
- Sparse Transformers
- Mixture of Experts (MoE)
- Long-context Models
- AI Agents
- Tiny LLMs

## **PHASE 12 — Tools & Libraries**

- NLTK
- spaCy

- Gensim
- PyTorch
- TensorFlow
- HuggingFace Transformers
- LangChain
- LlamaIndex

### **PHASE 13 — Projects**

Beginner:

- Spam Classifier
- Sentiment Analyzer

Intermediate:

- Chatbot
- Summarizer

Advanced:

- RAG Chatbot
- AI Research Assistant
- Multi-agent System

### **PHASE 14 — Best Learning Order**

1. NLP Foundations
2. Text Preprocessing
3. Traditional NLP
4. Embeddings
5. Deep Learning
6. Transformers
7. LLMs
8. RAG & Agents
9. Deployment

### **PHASE 15 — Best Resources**

- Stanford CS224N
- DeepLearning.AI NLP
- HuggingFace Course
- Kaggle

- Research Papers

### **FINAL NOTE**

This roadmap covers NLP from beginner to advanced GenAI/LLM engineering level and is designed for becoming an NLP Engineer, GenAI Engineer, or AI Researcher.